

Spring Microservices by Srinivas Dande

<u>Prerequisites:</u>	<u>Timings:</u>	7.30A.M to 9.30A.M (Mon - Fri)
Java8	<u>Course Duraion:</u>	3 Months (120 Hours)
Spring Framework	<u>Course Price:</u>	Rs. 20, 000
Hibernate Framework	<u>Trainer:</u>	Srinivas Dande
Maven		
MySQL		

Spring Microservices Course has three Sub Course

- 1) Spring Boot Course
- 2) MicroServices with Spring Boot and Spring Cloud Course
- 3) Spring Security and Spring OAuth2 Security Course

1) Spring Boot Course	Duration: 40 Hours
2) MicroServices with Spring Boot and Spring Cloud Course	Duration: 50 Hours
3) Spring Security and Spring OAuth2 Security Course	Duration: 30 Hours

1) Spring Boot Course

Duration: 40 Hours

Module 1 : Spring Data Access with Boot - Part 1**Objectives:**

In this Module, You will learn Data Access with Spring Boot Applications with JDBC , Data Sources and Various Spring Boot features.

Topics:

- Spring -JDBC Flow
- Developing Spring -JDBC Lab
- Understanding the Problems with this Lab.
- First Spring Boot Example
- Spring Boot Features.
- Starters
- Auto Configurations
- Exploring Starters
- Exploring Auto Configurations
- Exploring YAML Files
- Using Hikari DataSource
- Using Tomcat DataSource
- Using DBCP2
- Using DBCP
- Using JdbcTemplate
- Using NamedParameterJdbcTemplate
- Using Logging with Boot
- Understanding Spring Boot Start-Up Tasks
- Exploring CommandLineRunner
- Exploring @SpringBootApplication
- Exploring @ComponentScan

Module 2 : Spring Data Access with Boot - Part 2**Objectives:**

In this Module, You will learn Data Access with Spring Boot Apps with Hibernate, JPA

Topics:

- Spring Boot with Hibernate
- Spring Boot with JPA
- Spring Boot with Transactions

Module 3 : Spring Data Access with Boot - Part 3

Objectives:

In this Module, You will learn Data Access with Spring Boot Applications with Spring Data JPA , Spring Data Mongo

Topics:

- Spring Data JPA
- Spring Data Mongo
- Spring Boot with Data JPA
- Exploring CrudRepository
- CrudRepositoryMethods
- Exploring JpaRepository
- JpaRepository Methods
- Exploring MongoRepository
- MongoRepository Methods
- Writing Methods using DSL
- Using @Query with JPA-QL
- Using @Query with Native Queries

Module 4: Spring MVC with Boot

Objectives:

In this Module, You will learn how to develop Spring Boot Applications with Spring MVC

Topics:

- Spring MVC Flow
- Developing Spring MVC Application with Spring Boot
- Creating WAR and Deploying to External Server
- Creating FAT JAR and Running with Embedded Container.
- Using BootStrap
- Using ThymeLeaf
- Running FAT Jar from Command Line
- Spring MVC -Boot -JPA CRUD Operations

Module 5 : Spring REST with Boot

Objectives:

In this Module, You will learn Developing, Documenting and Testing REST API and Spring Boot and Swagger

Topics:

- Introduction to REST Web Services
- Understanding HttpRequest
- Request Headers
- Request Body
- Understanding HttpResponse
- Response Headers
- Response Body
- REST Flow with Diagram
- Controller Vs RestController
- @RequestBody Vs @ResponseBody
- @RequestBody Vs @PathVariable
- Developing REST Lab with Boot
- Understanding URL's
- Writing @GetMapping
- Writing @RequestMapping
- Controller Vs RestController
- @RequestBody Vs @ResponseBody
- @RequestBody Vs @PathVariable
- Developing Spring REST and Data JPA Lab with Boot
- Writing @GetMapping
- Writing @PostMapping
- Writing @PutMapping
- Writing @DeleteMapping
- Using @RequestBody and @PathVariable
- Using RestTemplate
- Using ResponseEntity
- Handling Exceptions
- Handling Exceptions Locally in Controller
- Using ResponseStatusException
- Handling Exceptions Globally in Centralized Place
- Writing Custom Messages
- Using ResponseEntity
- Working with Swagger
- Documenting REST API with Swagger
- Testing REST API with Swagger

Module 6 : Actuators and Boot Admin

Objectives:

In this Module, You will learn Boot features Actuators , Boot Admin Client and Server

Topics:

- Understanding Importance of Actuators
- Enabling Actuators in Spring Boot Application
- Understanding Actuators Endpoints
- Enabling and Disabling Actuators Endpoints
- Customizing Info Endpoint
- Customizing Logging Endpoint
- Customizing Metrics Endpoint
- Customizing Health Endpoint
- Writing Custom HealthIndicators
- Understanding Importance of Boot Admin Client and Server
- Setting Up Boot Admin Server
- Enabling Boot Admin Client in Application
- Monitoring Operational Info at Boot Admin Server Console

Module 7 : Async Communication with RabbitMQ

Objectives:

In this Module, You will learn Async Communication with Rabbit MQ.

Topics:

- Understanding Synchronous Communication
- Understanding Asynchronous Communication
- Importance of Message Broker in Distributed Systems
- HTTP Vs AMQP
- Installing Erlang
- Installing RabbitMQ
- Checking RabbitMQ Management Console
- Developing RabbitMQ Application with Spring Boot.
- Developing Message Senders
- Using RabbitTemplate class
- Developing Message Listeners
- Using @RabbitListener
- Execution Flow of RabbitMQ Example

Module 8 : Async Communication with Kafka

Objectives:

In this Module, You will learn about Async Communication with Kafka.

Topics:

- Understanding Synchronous Communication
- Understanding Asynchronous Communication
- Installing Kafka and ZooKeeper
- Understanding Kafka and ZooKeeper
- Using Kafka in Spring Boot Application.
- Developing Message Senders
- Using KafkaTemplate class
- Developing Message Listeners
- Using @KafkaListener
- Execution Flow of Kafka Example

Module 9 : Unit Testing Spring Boot Applications

Objectives:

In this Module, You will learn about Unit Testing Spring Boot Applications.

Topics:

- Create a Spring Boot App for Testing
- Writing Unit Tests
- Writing Integration Tests

2) MicroServices with Spring Boot and Spring Cloud Course

Duration: 50 Hours

Module 1: Introduction to MicroServices**Objectives:**

In this Module, You will learn about Monolithic Applications vs MicroServices, BookStore Case Study

Topics:

- Monolithic Architecture
- Benefits of Monolithic Architecture:
- Challenges with Monolithic Applications:
- What are MicroServices?
- MicroServices benefits
- Challenges with MicroService Architecture
- Number Of Endpoints for a MicroService
- Communication styles
- Synchronous Communication
- ASynchronous Communication
- MicroServices early adopters
- Case Study : JLCBookStore
- Understanding JLCBookStore Use-cases.
- Designing Various MicroServices for JLCBookStore
 - a. BookPriceMS
 - b. BookSearchMS
 - c. PlaceOrderMS
 - d. UserRatingMS
 - e. BookStoreWeb
- Communication among MicroServices of JLCBookStore

Module 2 : Developing BookStore MS ehco system**Objectives:**

In this Module, You will develop the 5 MicroServices designed from BookStore CaseStudy with Spring MVC, Spring REST , Spring Data JPA , Swagger

Topics:

- Developing BookPriceMS
- Step By Step Instructions to Develop BookPriceMS

- Write Code for BookPriceMS
- Using Swagger with BookPriceMS
- Testing with Swagger
- Developing BookSearchMS
- Step By Step Instructions to Develop BookSearchMS
- Write Code for BookSearchMS
- Using Swagger with BookSearchMS
- Testing with Swagger
- Developing PlaceOrderMS
- Step By Step Instructions to Develop PlaceOrderMS
- Write Code for PlaceOrderMS
- Using Swagger with PlaceOrderMS
- Testing with Swagger
- Developing UserRatingMS
- Step By Step Instructions to Develop UserRatingMS
- Write Code for UserRatingMS
- Using Swagger with UserRatingMS
- Testing with Swagger
- Developing BookStoreWeb
- Step By Step Instructions to Develop BookStoreWeb
- Write Code for BookStoreWeb

Module 3 : RabbitMQ , Feign with BookStore

Objectives:

In this Module, You will update the 5 MicroServices developed with RabbitMQ, Feign.

Topics:

- Understanding Async Use Cases in BookStore
- Step By Step Instructions to Use RabbitMQ
- Using Rabbit MQ with BookSearchMS
- Using Rabbit MQ with PlaceOrderMS
- Using Rabbit MQ with UserRatingMS
- Using Rabbit MQ with BookStoreWeb
- Understanding REST Use Cases in BookStore
- Understanding Feign
- Understanding Eureka
- Replacing RestTemplate Calls with Feign
- Step By Step Instructions to Use Feign
- Using Feign with BookSearchMS
- Using Feign with BookStoreWeb

Module 4 : Actuators , Boot Admin and Eureka with BookStore

Objectives:

In this Module, You will update the 5 MicroServices developed with Actuators , Boot Admin and Eureka

Topics:

- BookStore with Actuators
- Step By Step Instructions to Use Actuators
- Using Actuators with BookPriceMS
- Using Actuators with BookSearchMS
- Using Actuators with PlaceOrderMS
- Using Actuators with UserRatingMS
- Using Actuators with BookStoreWeb
- BookStore with Boot Admin
- Step By Step Instructions to Setup Boot Admin Server
- Using Boot Admin Client with BookPriceMS
- Using Boot Admin Client with BookSearchMS
- Using Boot Admin Client with PlaceOrderMS
- Using Boot Admin Client with UserRatingMS
- Using Boot Admin Client with BookStoreWeb
- BookStore with Eureka
- Step By Step Instructions to Setup Eureka Server
- Using Eureka Client with BookPriceMS
- Using Eureka Client with BookSearchMS
- Using Eureka Client with PlaceOrderMS
- Using Eureka Client with UserRatingMS
- Using Eureka Client with BookStoreWeb

Module 5 : ,Ribbon and Spring Cloud Load Balancer with BookStore

Objectives:

In this Module, You will update the 5 MicroServices developed with Ribbon or Spring Cloud Load Balancer

Topics:

- Exploring Feign, Eureka ,Ribbon and Spring Cloud Load Balancer
- Ways to Invoke one MS from other MS
- Using RestTemplate
- Using Feign
- Using Discovery Client

- Using Ribbon backed RestTemplate
- Using LoadBalancerClient

Module 6 : Hystrix, ELK Stack & Zuul with BookStore

Objectives:

In this Module, You will learn Hystrix, ELK Stack & Zuul nad update the 5 MicroServices developed with them

Topics:

- Exploring Hystrix
- Using Hystrix with BookStore MicroServices
- Exploring ELK Stack
- Using ELK Stack with BookStore MicroServices
- Exploring Zuul
- Using Zuul with BookStore MicroServices

Module 7: Cloud Config Client & Server and Cloud Bus with BookStore

Objectives:

In this Module, You will learn Cloud Config Client & Server and Cloud Bus and update the BookStore MicroServices echo system with them.

Topics:

- How to Configure Application Configuration Data
- Centralizing Application Configuration Data
- Introduction to Spring Cloud Config Server
- Using GIT with Spring Cloud Config Server
- Setup the Initial Project
- Setup GIT Repository
- Create the Property Files in GIT Repository
- Setup Config Server
- Setup Config Client with BookSearchMS
- Setup Config Client with BookStoreWeb
- Scaling the BookSearchMS to 3 Instances
- Change the Configuration Properties
- Using refresh end-point of actuator
- Understanding Cloud Bus
- Setup Cloud Bus
- Cloud Bus - Flow with Diagram
- Understanding Multiple Project Environments



- Development Environment
- QA Environment
- Staging Environment
- Production Environment
- Creating Different Property files for Different Environments
- Creating Different Profiles for Different Environments
- Enabling Required Profile
- `spring.profiles.active=dev`

Module 8 : Kafka with BookStore

Objectives:

In this Module, You will update the 5 MicroServices developed with Kafka by replaing Rabbit MQ

Topics:

- Understanding Async Use Cases in BookStore
- Step By Step Instructions to Use Kafka
- Using Kafka with BookSearchMS
- Using Kafka with PlaceOrderMS
- Using Kafka with UserRatingMS
- Using Kafka with BookStoreWeb

3) Spring Security and Spring OAuth2 Security Course

Duration: 30 Hours

Module 1 : Spring Security with Boot**Objectives:**

In this Module, You will learn applying Authentication and Authorization for Spring Boot Applications with Spring Security Framework.

Topics:

- Configuring Default Spring Security
- Exploring Security Filters and Filter Chains
- Security Processing Flow
- Configuring Authentication Managers
- In-memory Authentication Manager
- JDBC Authentication Manager
- LDAP Authentication Manager
- Configuring HttpSecurity for Authorization
- Denying all the requests
- Permit all the requests
- Permit for specific requests
- Customizing Login page
- Customizing Logout page
- User Registration
- Password Encoders
- Displaying Links as Per the Roles
- Exploring CORS and CSRF
- Exploring JWT

Module 2: Spring OAuth2 Security**Objectives:**

In this Module, You will learn about Spring OAuth2 Security with All Grant Types and Writing OAuth2 Clients.

Topics:

- Understanding OAuth2 Framework
- Understanding how OAuth2 allows applications to obtain limited access to resources
- Main components of the OAuth 2.0 specification



- Understanding Resource Owner
- Understanding Client Application
- Understanding Authorization Server
- Understanding Resource Server
- Exploring OAuth2 endpoints
- OAuth2 Grant Types
- Exploring Authorization Code Grant Type
- Exploring Implicit Grant Type
- Exploring Resource owner password credentials Grant Type
- Exploring Client credentials Grant Type
- Refresh Token Grant Type
- Choosing Required Grant Type
- OAuth2 Client Applications
- OAuth2 Client for Authorization Code Grant Type
- OAuth2 Client for RO Password Grant Type
- OAuth2 Client for Implicit Grant Type
- OAuth2 Client for Client Credentials Grant Type
- Token based Authentication with JWT
- OAuth2 Complete Example

Module 3: MicroServices Security with OAuth2

Objectives:

In this Module, You will learn Applying Security and OAuth2 Security to BookStore MicroService Eco System.

Topics:

- Applying Spring Security and OAuth2 Security for following microservices of BookStore Echo System.
 - a. BookStoreWeb
 - b. BookSearchMS
 - c. BookPriceMS
 - d. PlaceOrderMS
 - e. UserRatingMS